



CHAPTER

Commissioning

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5.1 Before the first start

Confirm every hydraulic connection in section 4.4 is complete and leak-checked before energizing the cabinet.
TODO: no document describing a formal pre-start electrical isolation/lockout check or inspection checklist was found beyond the piping steps.

5.2 Filling and priming

Install the submersible feed pump in the source water body; connect its outlet to the pretreatment 2½" (F) inlet with 60 mm PVCu pipe; connect the pretreatment filtered-water outlet to the treatment container inlet with 40 mm PVCu pipe; join both containers' drain outlets via a tee to the drain tank; verify the product-water outlet is connected per site requirements. TODO: no document states specific priming steps (venting, air removal, initial fill sequence) beyond making these connections.

5.3 First power-up

TODO: no formal first-power-up procedure (breaker/contactor closing order, PLC boot checks, HMI startup) was found. What *does* exist is a real commissioning engineer's log from a pilot startup, which illustrates the general shape of a ramp-up sequence (source: pilot commissioning notes, HydroVolta engineering library) — treat it as an informative example, not a validated procedure for the standard product:

1. Start the feed and process pumps at a low VFD frequency (~ 35 Hz in the logged example) to establish flow gently rather than at full speed.
2. Increase the dosing-pump VFD frequency in steps (e.g. to ~ 60 Hz) once feed flow is confirmed stable.
3. Confirm each stream reaches a stable pressure and flow reading before proceeding — the log explicitly checks “stable pressure” and “stable flow” at each stack position before applying stack current.
4. Apply DC current to the stack(s) only once hydraulics are stable, and confirm conductivity in the target stream stabilizes at the expected value.
5. Bring additional stacks online one at a time if the skid has more than one — the log notes that current draw on an already-running stack visibly drops as a newly started stack begins drawing current, which is expected stack-to-stack interaction, not a fault.

5.4 Setting the operating parameters

Dosing pumps are controlled via a 4–20 mA analog signal from ADAM I/O modules. Default configuration: **SET 1** = 4.0 mA (minimum, pump stopped, 0 impulses/min), **SET 2** = 20.0 mA (maximum, 180 impulses/min); internally, a raw integer of 0 corresponds to 3.8 mA and 4095 to 19.7 mA. Each dosing channel is verified across 4/8/12/16/20 mA. TODO: hydraulic setpoints (target flow/pressure) and stack-voltage setpoints are referenced in HydroVolta's internal user-manual outline but no actual values were found — see also appendix A.

5.5 Functional test

A real, filled-in functional test record exists for the BASF pilot's pretreatment dosing pumps and control panel:

1. Verify each dosing channel's SET 1/SET 2 defaults (4.0 mA / 0 impulses/min, 20.0 mA / 180 impulses/min) and confirm out-of-range signals (below 4 mA, above 20 mA) trigger a stop.
2. Test every alarm digital input and dosing digital output for correct read/operate behavior.
3. Drive each dosing pump across 4/8/12/16/20 mA and confirm the correct response at each step.

TODO: no document describes functional tests for the treatment (SonixED™/stack) side, pumps, or sensors beyond pretreatment dosing — see the relevant Part II chapters for what should be tested.

5.6 Handover

TODO: no document describing a formal handover procedure, sign-off sheet, or customer training checklist was found.

